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ORIGINAL PAPER

Modulating mouse innate immunity to RNA viruses by expressing the *Bos taurus* Mx system

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Abstract Mx proteins are interferon-induced members of the dynamin superfamily of large guanosine triphosphatases. These proteins have attracted much attention because some display antiviral activity against pathogenic RNA viruses, such as members of the orthomyxoviridae, bunyaviridae, and rhabdoviridae families. Among the diverse mammalian Mx proteins examined so far, we have recently demonstrated in vitro that the *Bos taurus* isoform 1 (boMx1) is endowed with exceptional anti-rabies-virus activity. This finding has prompted us to seek an appropriate in vivo model for confirming and evaluating gene therapy strategies. Using a BAC transgene, we have generated transgenic mouse lines expressing the antiviral boMx1 protein and boMx2 proteins under the control of their natural promoter and short- and long-range regulatory elements. Expressed boMx1 and boMx2 are correctly assembled, as deduced from mRNA sequencing and western blotting. Poly-I/C-subordinated expression of boMx1 was detected in various organs by immunohistochemistry, and transgenic lines were readily classified as high- or low-expression lines on the basis of tissue boMx1 concentrations measured by ELISA. Poly-I/C-induced Madin-Darby bovine kidney cells, bovine turbinate cells, and cultured cells from high-expression line of transgenic mice were found to contain about the same concentration of boMx1, suggesting that this protein is produced at near-physiological levels. Furthermore, insertion of the bovine Mx system rendered transgenic mice resistant to vesicular-stomatitis-virus-associated morbidity and mortality, and embryonic fibroblasts derived from high-expression transgenic mice were far less permissive to the virus. These results demonstrate that the *Bos taurus* Mx system is a powerful anti-VSV agent in vivo and suggest that the transgenic mouse lines generated here constitute a good model for studying in vivo the various antiviral functions—known and yet to be discovered—exerted by bovine Mx proteins, with priority emphasis on the antirabic function of boMx1.

Keywords Innate resistance · Virus · VSV · Host–pathogen

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Introduction

Living organisms are constantly mobilizing numerous and powerful defense mechanisms in order to resist pathogenic agents. A method of choice for identifying the molecular actors of these defense mechanisms is to detect and analyze variant organisms in which the defenses are altered. Although many variants have been clearly identified in plants (Fraser 1990) and mammals (Green 1989), notably by us (Faisca et al. 2005; Bui Tran Anh et al. 2006), only a few genes and their associated proteins have been characterized to date. One of the best-studied virus-disease susceptibility variants studied is the A2G mouse strain, with its exceptional resistance to the influenza virus. This innate resistance, discovered by Lindenmann (1962), concerns orthomyxoviruses specifically. It protects A2G mice effectively against high virus doses whatever the inoculation route and is inherited as a single autosomal dominant trait (reviewed in Haller 1981). It has now emerged that this resistance is caused by a single gene, named *Mx1*, whose transcription is induced by type-I interferons, themselves being induced by viral infection. The gene codes for a single GTPase, the Mx1 protein (reviewed in Haller et al. 2007). The Mx1-dependent susceptibility of most laboratory mouse lines is due to a structural modification of the *Mx1* gene, either a point substitution creating a premature stop codon or a deletion of exons 9–11. This leads to synthesis of a C-terminally-truncated Mx1 protein (Staeheli et al. 1988). Since this discovery it has emerged that all vertebrates possess between two and three interferon-inducible *Mx-like* genes and that vertebrate Mx proteins, while targeting RNA viruses as a rule, have different antiviral spectra. From these observations has stemmed the hypothesis that each Mx protein might inhibit a set of viruses specifically pathogenic towards the species in which it evolved. One might thus assume that within the pool of Mx isoforms available in nature, there exist diverse antiviral proteins likely to reveal as yet unknown antiviral mechanisms and thus to promote the development of new antiviral drugs. Alternatively, Mx-encoding polynucleotides might be exploitable in gene therapy.

Previously we looked closely at the Mx proteins of the bovine species and we validated a cell culture system for evaluating the antiviral spectrum of the Mx1 protein (boMx1) in vitro (Gérardin et al. 2004; Baise et al. 2004; Leroy et al. 2005). This work revealed an antiviral activity directed against the vesicular stomatitis virus (VSV). In another in vitro study, we recently showed that the boMx1 protein can also very strongly repress two different rabies virus strains, although the antirabic activity of the prototype Mx protein (human MxA [huMxA]) is negligible (Leroy et al. 2006). As rabies is an incurable and inexorably fatal disease in humans, causing some 70,000 deaths annually (http://www.who.int/), the unique antirabic activity of boMx1 has quite understandably raised hopes of developing at last a first curative, rather than merely supportive, therapeutic strategy. Yet a prerequisite to such a project is the demonstration that the antirabic resistance conferred by boMx1 in vitro can be reproduced in vivo. The aim of the present work was to create and characterize transgenic mouse lines equipped with the bovine Mx system. As the three previous attempts to produce a transgenic animal expressing an exogenous Mx protein resulted in multiple problems of expression (Arnheiter et al. 1996), we have chosen to use a transgene of a totally different nature, which has enabled us to produce mouse lines endowed with a bovine Mx operon mimicking very closely the way the native system functions. These lines should be useful for examining in vivo the various antiviral functions—known and yet to be discovered—of bovine Mx proteins, with priority emphasis on the antirabic function of boMx1.

Materials and methods

Characterization of the bacterial artificial chromosome transgene

A single bacterial artificial chromosome (BAC) clone (305L8) unequivocally containing the whole *Bos taurus Mx1* gene (*boMx1*) was identified by screening the RPCI-42 BAC library made from Holstein bull white blood cell DNA (bacpac.chori.org). As DNA probes, we used 5′ (nt 281–315) and 3′ (nt 1,837–1,871) fragments of the *boMx1* gene (U88329) radioactively labeled with $[^{32}P]$dCTP. Following expansion, small amounts of clone 305L8 DNA were isolated by alkaline lysis, and preparative BAC DNA isolation was carried out with the Nucleobond AX kit (Macherey-Nagel). By *NotI* restriction digestion and pulse-field gel electrophoresis, we estimated the size of the genomic insert to be about 220 kb (data not shown). The DNA sequences of the genomic insert extremities were determined with the Universal Genome Walker™ kit (BD Bioscience Clontech). Pools of adaptor-ligated genomic DNA fragments of the BAC were constructed by digesting BAC DNA with a set of restriction enzymes. The two fragments containing the vector-insert interfaces were PCR amplified with the help of adaptor- and pBACe3.6-specific primers (Frengen et al. 1999), the latter two corresponding to the regions directly flanking the insert (the T7 and Sp6 promoters). Nucleotide sequencing of the PCR bands revealed the BAC end sequences, which could be anchored on the *Bos taurus* genome sequence (www.ensembl.org). The BAC genomic insert was found to extend over 211,270 bp and to include the complete *boMx2* (AF355147) and *boMx1* (AF047692) genes and the truncated coding sequence of the androgen-regulated serine protease *boTMPRSS2* (BC133425). Further comparison of the BAC sequence with the human genome sequence revealed the putative presence of the bovine homologue of *FAM3B* (NM_058186).

Generation of transgenic mice carrying *BAC305L8*

The $Mx1^{-/-}$ allelic status of FVB/J mice at the *Mx1* locus was first demonstrated by combining in silico comparisons of available SNP data from a series of strains (http://www.informatics.jax.org/), PCR amplification of the intron 10 to exon 11 junction and PCR-RFLV analysis of exon 14 using *Hha*I as described (Jin et al. 1998; Vanlaere et al. 2008). The purified BAC DNA was dissolved in microinjection buffer (10 mM Tris–HCl [pH 7.5], 0.1 mM EDTA, 30 μM spermine, 70 μM spermidine, 100 mM NaCl) at a concentration of about 2–4 ng/μl and microinjected into the pronuclei of FVB/J blastocysts. These were subsequently implanted in pseudo-pregnant recipients. Screening for integration of the *BAC305L8* transgene in the resulting offspring and testing for further germ line transmission (after crossing of selected animals with wild-type FVB/J) were done by DNA genotyping following PCR with *Bos taurus*-specific *Mx1* primers (Table 1). About 11 of 350 offspring contained at least one copy of the transgene, of which nine proved to transmit it to the next generation. Hemi- and homozygous transgenic mice were obtained through further breeding. F6 animals were analyzed for expression and functional studies of the inserted bovine *Mx* genes. Construction of these mice was authorized by the University Bioethics Committee, and anesthesia/euthanasia procedures were consistent with the recommendations of the American Veterinary Medical Association. All mice were housed in a temperature-controlled room (22°C) with 12 h light/12 h dark cycling and fed Purina or Altromin chow.

Generation of embryonic fibroblasts carrying *BAC305L8*

Primary mouse embryonic fibroblasts (MEF) from wild-type and transgenic mice were harvested from 14-day-post-coitum embryos. First the head, liver, and intestine were dissected and the remaining fetal tissues were minced and rinsed in PBS. Fetal homogenates were then treated with trypsin (0.25% in Dulbecco's PBS), incubated for 30 min at 37°C, and subsequently dissociated in medium. After removal of perceptible tissue clumps, the remaining cells were plated out in a 25-cm^2 flask containing DMEM supplemented with 10% heat-inactivated FCS, 1% (v/v) penicillin-streptomycin, and 0.5% amphotericin B. After a 4 h incubation, nonadherent cells were eliminated by gentle mixing, directly followed by medium replacement. Primary cultures reached confluence after ~60 h and were split 1:2 for freezing in liquid nitrogen (passage 1 MEFs) or for plating out in 175-cm^2 flasks. For semi-continuous culturing, MEF cultures were split 1:4 approximately every 4 days.

Analysis of *boMx* mRNA levels

Mx transcript levels were compared in wild-type and transgenic mice, after a standardized stimulation (poly-I/C, 15 μg/kg ip for 24 h). Amounts of boMx-protein-encoding mRNAs were normalized with respect to the amount of endogenous reference mRNA (encoding glyceraldehyde-3-phosphate dehydrogenase, GAPDH).

Production of cDNA samples

Brain, lung, and spleen tissues (50–100 mg) were taken from wild-type and transgenic 7- to 8-week-old female poly-I/C-stimulated mice (five specimens of each line tested). Each tissue sample was individually homogenized (Qiagen's TissueLyser, 30 Hz for 5 min) in TRIzol (Invitrogen) for preparation of total mRNA. Each homogenate was treated with TURBO DNase

Table 1 Primers used in genotype (PCR) and expression (real time PCR) analyses

Target	Amplified fragment length (bp)	Primer sequences
Genotyping		
boMx1-upstream	1,100	Fwd: 5′-CATTCTCTTGATTGGGGAGCTTTA-3′
		Rev: 5′-GCTTCAAAATTCACATTATGCTCA-3′
boMx1-downstream	150	Fwd: 5′-CCGTAGTCTCTGCTGTCTCT-3′
		Rev: 5′-ACCCTTCTACAGTGCTGTGT-3′
boMx2-upstream	883	Fwd: 5′-AATTTGCCCACAAGTCAGG-3′
		Rev: 5′-AGACTCGAGAGCCACGTTTATCAGGAAGC-3′
boMx2-downstream	824	Fwd: 5′-TTGGTTGGTACTGACCACTG-3′
		Rev: 5′-AACTGGATTTAAGCCACAGC-3′
moMST	350	Fwd: 5′-AGTGGAAGAAATTCTCTCTTCACTC-3′
		Rev: 5′-GTAGCTCACCTCACCCTGCATGTT-3′
moMx1[IN10/EX11]	290	Fwd: 5′-GTGACCTTTGAACCTGCTTCCT-3′
		Rev: 5′-GCAGACTCTTCCAGGGCTTTGA-3′
moMx1[EX3/IN3/EX4]	1,100	Fwd: 5′-GTATTGACCTCATCGACACCCT-3′
		Rev: 5′-GGGCATCTGGTAACAATACCTA-3′
moMx1[EX14]	334	Fwd: 5′-CCGTTGCTCATCTCCGACTGT-3′
		Rev: 5′-CACCTTCCTCTGCCCCTACCT-3′
mRNA expression analysis		
moGAPDH	156	Fwd: 5′-TGGCAAAGTGGAGATTGTTGCC-3′
		Rev: 5′-AAGATGGTGATGGGCTTCCCG-3′
boMx1	202	Fwd: 5′-GGGAATGAAGACGAGTGGAA-3′
		Rev: 5′-TGCCAGGAAGGTCTATCAGG-3′
boMx2	201	Fwd: 5′-CAGAAGGCCATGGAAATTGT-3′
		Rev: 5′-CACGCCGTAAATCTGGTCTT-3′

Bo Bovine, *mo* mouse, *MST* myostatin, *GAPDH* glyceraldehyde-3-phosphate dehydrogenase, *IN* intron, *EX* exon

(Ambion) for 30 min at 37°C. Next, line-and organ-specific total RNA extracts were produced by pooling the five individual extracts. After purification the purity and concentration of each extract were determined spectrophotometrically (the OD260/280 and OD260/230, respectively, were in the range 1.9 → 2.0 and 1.8 → 2.2, NanoDrop-1000/Isogen), and mRNA integrity was checked by agarose gel electrophoresis. An aliquot of each line- and organ-specific total RNA extract (2 μg RNA) was then reverse-transcribed at 50°C for 60 min in the presence of 50 μg 18-mer oligonucleotide primers (Table 1) and Superscript III reverse transcriptase (Invitrogen).

Production of DNA calibrators

To estimate the number of copies of each target cDNA in each sample, it was necessary to generate calibrators containing known target copy numbers. To this end, each target sequence was amplified by RT-PCR (see primers in Table 1), purified (Nucleospin Extract II, Macherey-Nagel), and cloned into the pCRII vector (Invitrogen). Then stock solutions of known concentration were generated for each target, the number of copies (per ml) being calculated by dividing the plasmid concentration (μg/μl) by the mass of the plasmid (μg). For this calculation the plasmid concentration was determined by spectrometry (OD at 260 nm) and the plasmid mass was calculated by multiplying the length of the plasmid (vector length [bp] + insert length [bp]) by the mass of a nucleotide ($1.096\ 10^{-15}$ [μg]). For each target, a standard curve for quantitative PCR was constructed on the basis of six dilutions of the appropriate stock solution, corresponding to 5×10^1, 5×10^2, 5×10^3, 5×10^4, 5×10^5, and 7.5×10^5 copies. For comparisons between lines and organs, the number of *Mx* transcripts was normalized with respect to the corresponding number of *GAPDH* transcripts.

Real-time PCR

The primer pairs used to detect fragments of the *boMx1* and *boMx2* transgene transcripts and of the mouse *GAPDH* transcript are listed in Table 1. The PCR mixture consisted of 100 ng/µl template DNA (1 µl), 70 nM primers (0.7 µl of each), and 12.5 µl ABsolute^{TM} Blue QPCR SYBR Green ROX Mix (ABgene) in a final volume of 25 µl. The mixture was placed in an ABI PRISM® 7900HT thermocycler and amplification was carried out under the following conditions: initial denaturation at 95°C for 15 min, followed by 40 cycles of denaturation at 95°C for 15 s and annealing-extension at 58°C for 30 s, and then a final extension at 72°C for 30 s. Amplification of all transcripts was performed in triplicate, and three independent sessions were carried out with each RNA extract. The melting curve of each amplicon was monitored by means of a swing back to 50°C, followed by a stepwise rise in temperature up to 95°C. Melting curve analysis always revealed the presence of a single product. To check for false positives, RT-free and no-template controls were run for each template and primer pair.

Analysis of boMx1 protein levels

Production of antisera targeting Bos taurus Mx proteins

A 1,974-bp fragment of the *boMx1* gene (NM_173940.2) was PCR amplified from template DNA extracted from IFNα-induced (1,000 U/ml recombinant IFNα A/D for 24 h, Sigma) Madin-Darby bovine kidney cells (MDBK, ATCC # CCL-22) as previously described (Baise et al. 2004). The forward primer was 5′-ggg**ggatcc**gatggttcattctgacttgggtatcg-3′, and the reverse primer was 5′-ccc**aagctt**gcccgggaacttgg ccagc-3′, with the boldface type indicating the *BamHI* and *HindIII* restriction sites chosen for cloning in the vector pET-28b[+] (Novagen). A PCR fragment of the expected size was then generated and cloned. The presence and integrity of the recombinant *boMx1*-pET28b plasmid were confirmed in the kanamycin-resistant transformants by restriction endonuclease digestion and by sequencing of the *boMx1* cDNA. The recombinant plasmid was used to transform competent *E. coli* Rosetta DE3 pLysS cells (Novagen), which were induced with 1 mM isopropylthio-ß-D-galactoside for 2 h and then harvested by centrifugation (1,000 g for 10 min at 4°C). The pellets were stored at −80°C. Total protein extraction and purification under denaturing conditions were done with and as recommended for the ProBond Purification system (Invitrogen). Elution fractions showing a measurable absorbance at 280 nm and a band at ~80 kDa on western blots (corresponding to recombinant boMx1) were pooled, supplemented with 0.5% Triton-X100, and dialyzed stepwise against successive dilutions of PBS-urea (6–0.5 M). The concentration of recombinant boMx1 in the resulting stock solution was determined with the BCA Protein Assay Kit (Pierce Biotechnology), and a rabbit antiserum was obtained from two rabbits injected four times ip with boluses containing 200 µg boMx1. A second antiserum was raised in FVB/J mice by six consecutive ip injections with 100 µl boluses containing spleen extracts from poly-I/C-induced (15 µg/g body weight 24 h before spleen extraction) transgenic mice.

Extraction

For each mouse line and each tissue sampled (brain, heart, kidney, liver, lung, and spleen) the frozen (−80°C) organs from five poly-I/C-exposed (15 µg/g body weight 24 h before sacrifice) 8-week-old animals were pooled and pulverized with pestle and mortar in a liquid nitrogen bath. About 500 mg crude frozen homogenate of each organ was resuspended in 600 µl extraction buffer (LLB, Eurogentec), vortexed for 2 min, and kept on ice. Samples were then sonicated (three pulses of 30 s each with 30-s intervals on ice) and centrifuged at 11,000*g* for 10 min at 4°C. The supernatants were stored at −80°C in protein-repellant-coated tubes (Protein LoBind Eppendorf Tubes®) and total protein content was measured with the BCA Protein Assay kit.

Immunoblot analysis

Aliquots of resulting supernatants corresponding to 50 (spleen), 70 (lung), or 250 µg (brain) total protein were loaded onto 10% SDS–PAGE gels and electrophoresed.

After electrotransfer onto nitrocellulose membranes, the blots were blocked for 30 min with Tris-buffered saline containing 0.1% Tween and 10% bovine serum albumin and incubated for 1 h at room temperature with the mouse antiserum (dilution 1:1000). Immune complexes were revealed with HRP-conjugated pig anti-rabbit IgG F(ab′)2 fragments (dilution 1:1000, Dakocytomation), and peroxidase detection with the CN/DAB Substrate Kit (Pierce Biotechnology). Densitometry was performed with the Fluor S Multiimager CCD camera system and Quantity One software (Bio-Rad).

ELISA

A non-competitive indirect sandwich enzyme-linked immunosorbent assay (ELISA) was developed, using rabbit antiserum for capture, mouse antiserum followed by HRP-conjugated polyclonal rabbit-anti-mouse-Ig (#P0260, Dakocytomation) for detection, and finally TMB conversion as the read-out parameter for enzyme activity (Enhanced K-Blue, Neogen). First, 96-well Microlon 600 plates (Greiner, #655081) were coated overnight at 4°C with 100 μl rabbit antiserum diluted 1:1000 in carbonate buffer (100 mM, pH 9.5). They were then blocked for 1 h at 37°C with 250 μl casein solution (1% in PBS). For boMx1 content determinations, the wells were first incubated for 1 h at 37°C with 100 μl calibrator (see below) or with organ extract diluted in PBS, then incubated for 1 h at 37°C with 100 μl mouse antiserum diluted 1:1000 in PBS-0.5% casein. For detection of immune complexes, the wells were incubated for 1 h at 37°C with 100 μl rabbit anti-mouse-Ig diluted 1:1000 in PBS-0.5% casein, then incubated at room temperature for 20 min with 100 μl TMB in substrate buffer with H_2O_2, according to the manufacturer's recommendations. Development was stopped by adding 100 μl of 1 M HCl and the plates were read at 450 nm. The OD was determined with respect to a subtractive reference (an extract of the corresponding organ from stimulated wild-type FVB/J mice). Calibrators were derived from a stock of recombinant boMx1 titrating 100 μg/ml. The highest calibrating concentration used was 375 ng/ml, and this solution was subjected to a 9-step serial dilution in PBS. The concentrations of the resulting calibrating samples were as follows: 375, 250, 200, 150, 100, 80, 60, 40, 20, and 10 ng/ml. New calibrators were generated from the stock for each session of tissue boMx1 content measurements. They provided an absolute correlation of signal vs. concentration (ng boMx1 per μg soluble protein). Successive dilutions of primordial protein extracts from each organ were first assayed in order to determine the range of concentrations yielding the highest signal. Spleen, lung, and brain protein extracts were diluted so as to incorporate respectively ~5, ~50, and ~300 μg total protein per well.

Immunohistochemistry

Tissue sampling was performed according to a standard protocol. After fixation in 4% neutral-buffered ice-cold paraformaldehyde and embedding in paraffin, tissue sections were stained for boMx1 detection by an indirect immunohistological method using the rabbit antiserum followed by an HRP-conjugated goat-anti-rabbit immunoglobulins secondary antibody. Peroxidase was revealed with 3-amino-9-ethyl-carbazole, resulting in a bright red precipitate. Tissues were counterstained with Mayer's hematoxylin and embedded in glycerol–gelatin. Rabbit pre-immunization serum, omission sections, and mock-exposed MDBK cell cytospins were used as negative controls and IFNα-exposed MDBK cell cytospins were used as positive controls. The following tissues were examined: lung, heart, intestine, liver, spleen, kidney, cerebrum, cerebellum, and brain stem.

Functional analysis of boMx proteins

Modulation of mouse innate immunity against RNA viruses by the transgene products was probed by examining whether the biological cycle of the vesicular stomatitis virus (VSV), a rhabdovirus, is altered in transgenic MEFs or mice. A stock and appropriate dilutions of VSV serotype Indiana were prepared from the supernatant of virus-infected BHK-21 cells. Two independent in vitro experiments were first conducted, in which viral suspensions were incorporated (300 μl/well, 24-well plates) for 1 h at a multiplicity of infection of 0.1, 1, or 10 into near-confluent (80%) cultures of noninduced or induced (50 μg/ml poly-I/C for 24 h) MEF lines derived from wild-type (WT), transgenic low-expression, and transgenic high-expression FVB/J mice. After a 1 h adsorption period, excess inoculum was removed by washing with PBS, and the cultures were re-incubated for 24 h at 37°C in fresh DMEM. The culture supernatants were then sampled for virus titration. For in vivo studies, sets of 15 WT, 10 ML-555, and 15 ML-549 FVB/J mice were inoculated with the virus by slowly instilling 50 µl of the viral suspension (i.e., $\sim 10^7$ cell culture infective dose 50% [$CCID_{50}$]) into the nostrils under anesthesia (30/5 mg kg^{-1} xylazine/ketamine ip). In all sets, body weight and survival were monitored for 14 days. Subsets of 5 WT and 5 ML-549 mice were euthanized on day 4 after inoculation. Their lungs and brains were removed and homogenized with a TissueLyser for subsequent virus titration. Viral titers from supernatants and organ suspensions were first determined in duplicate on Vero cells. They were expressed in $CCID_{50}$ units at 48 h after inoculation as previously described (Baise et al. 2004). Relative quantification of the viral load was also done by real-time PCR. Total RNA was extracted with the help of commercially available NucleoSpin silica-based spin-columns according to the manufacturer's instructions (Macherey-Nagel). In a separate reverse transcription (RT) step, 2 µg extracted RNA was added to 1× Multiscribe RT Buffer (TaqMan® Reverse Transcription Reagents, Applied Biosystems) supplemented with 25 pmol random hexamer primers (Applied Biosystems), 5 nmol dNTP's, 55 nmol $MgCl_2$, 4 IU RNase inhibitor; and 12.5 IU Multiscribe reverse transcriptase (Applied Biosystems) in a total volume of 10 µl. RT conditions were as follows: 10 min at 25°C, followed by 30 min at 48°C and 5 min at 95°C. For the subsequent real-time PCR, 5 µl template cDNA was added to 12.5 µl of 2× SYBR Green PCR Master Mix (Applied Biosystems) supplemented with 5 pmol forward and reverse primers in a total volume of 25 µl (for primer sequences, refer to Núñez et al. 1998). The mixture was placed in an ABI 7900HT thermocycler for 10 min at 95°C, then the targeted VSV-specific cDNA segment was amplified by means of a program consisting of 40 cycles of 15 s at 95°C and 60 s at 60°C. The melting curve of the resulting amplicon was monitored by means of a swing back to 50°C for 15 s, followed by a stepwise rise in temperature up to 95°C. A VSV-positive sample and a negative water sample were included as internal controls in both the RT and the PCR step. All samples were analyzed in duplicate reactions.

Statistical analysis

All data are reported as means ± SD. Differences between mean values were tested for statistical significance by the two-tailed Student's *t* test, survival curves were subjected to Kaplan-Meier analysis, and the significance threshold was set at $P < 0.05$.

Results

Development of transgenic mouse lines with a functional *BAC305L8* insert

SNP data available between positions 976 686 42 and 976 845 14 on chromosome 16 (*Mx1* gene) do not reveal variation between the FVB/J line on the one hand, BALB/c, C57BL/6, DBA/2, and C3H/HeN lines on the other. Moreover, the genomic stretch overlapping intron 10 and exon 11 was retrieved by PCR from BALB/c-A2G mice but never from BALB/c, FVB/J and ML-549 lines (Fig. 1A). PCR-RFLV analysis of exon 14 revealed the presence of an *Hha*I restriction site in all strains tested, which refuted the hypothesis of a CBA/J-like $Mx1^{-/-}$ allele in FVB/J (Fig. 1B). Collectively, these results show that the FVB/J line carries the Mx1-negative allele common to the vast majority of inbred lines. About 11 mice born through oviduct transfer of microinjected oocytes were transgenic. Nine of these transmitted the transgene to their offspring, as revealed by PCR analysis, but two of the nine lines became rapidly extinct because of very low reproduction rates. Next, a more detailed analysis by standard PCR enabled us to amplify from the DNA of the seven remaining lines specific segments corresponding to the 5′ and 3′ ends of the *boMx1* and *boMx2* genes. This suggests that at least one intact copy of each bovine *Mx* gene was inserted in each line (Fig. 1).

Maximum levels of *boMx1* and *boMx2* mRNAs were measured in lung tissues from poly-I/C-injected mice by reverse transcription followed by real-time PCR. They were normalized against GAPDH mRNA. This analysis demonstrated efficient transcription of the inserted *boMx1* and *boMx2* genes in the lungs in all lines except ML-555, where *boMx2* mRNA was never detected (Fig. 1). The transgenic lines were readily classified as high-expression (ML-549, ML-556), medium-expression (ML-310, ML-375), or

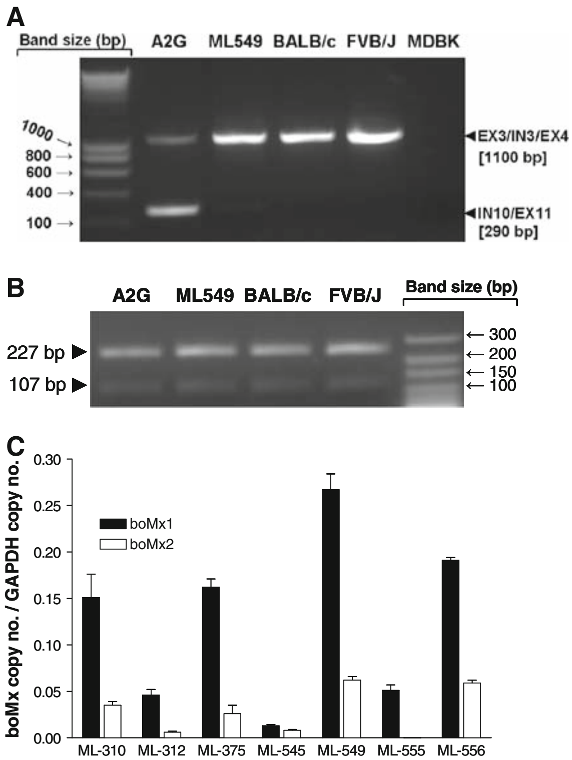


Fig. 1 **A** and **B** Genotyping of the *moMx1* locus in mice with known (BALB/c-A2G, BALB/c) and unknown (FVB/J and ML-549) allelic status. The BALB/c-A2G congenic line was generously provided by Drs O. Haller and P. Staeheli. **A** FVB/J and ML-549 mice carry the BALB/c-like *moMx*$^{-/-}$ allele. Targeted genomic segment consists in the junction between intron 10 and exon 11 which are deleted in most of inbred mouse strains but not in A2G (Staeheli et al. 1988). Primers are listed in Table 1. **B** The *moMx1*$^{+/+}$ allele of BALB/c-A2G mice and *Mx1*$^{-/-}$ alleles of BALB/c, FVB/J and ML-549 carry the *Hha*I restriction site in exon 14 (two *bands*), which suggests that the *moMx*$^{-/-}$ allele of FVB/J is that of BALB/c rather than that of CBA/J (Staeheli et al. 1988). Primers are listed in Table 1. **C** Detection and quantitation by real-time PCR of *boMx1* and *boMx2* transcripts in lung extracts from transgenic mice. Induction was carried out with poly-I/*C* for 24 h before sampling. Using a set of lungs from specific pathogen-free noninduced mice, the positive/negative cut-off threshold was set at a Ct value of 28. *GAPDH* transcript copy number was used for normalization. See text for key

low-expression lines (ML-312, ML-545 & ML-555), according to the amount of transcripts produced. In silico translation of the boMx polynucleotide sequences retrieved by RT-PCR from poly-I/C-induced transgenic mice yielded the expected amino acid sequences.

Western blot (immunoblot) analysis of lung (Fig. 2B), spleen, and brain (data not shown) extracts showed that *boMx1* mRNA was duly translated in all three tissues. A semi-quantitative densitometric analysis of three blots, obtained from the lungs of three mice, yielded the same classification by expression level as determined by real-time RT-PCR: ML-549 ≈ ML-556 > ML-310 ≈ ML-375 > ML-312 ≈ ML-545 ≈ ML-555 (Fig. 2B). Immunohistochemical analysis of brain, heart, lung, intestine, kidney, liver, and spleen tissues confirmed these results and further revealed inter- and intra-organ differences in expression (Fig. 3). In all mice of all transgenic lines but one (ML-545, whose tissues displayed no staining), boMx1-specific staining was more intense in the kidneys and intestines than in the other organs. In the brain and liver, it was more intense in some cell types (Kupffer cells) or structures (the choroid plexus); in the lungs, staining was more intense in the epithelium of the alveoli than in the epithelium of the bronchioles. We used an ELISA to measure the concentration of boMx1 protein in various organs of five poly-I/C pretreated mice of each line. For line ML-545, whatever the animal or the organ, the results were the same as for the organs of wild-type mice. Among the six boMx1-producing lines, no clear expression pattern emerged, although the dominant trend was for the concentration to be about 15 times as high in the spleen as in the lungs and about five times as high in the lungs as in the brain (Fig. 2C). In the lungs, four lines displayed concentrations of 200–300 and the other two about 100 ng per mg soluble protein. On the basis of concentrations in the brain, three pairs were identifiable: ML-549 and ML-556 with about 50, ML-310 and ML-375 with 15-25, and ML-312 and ML-375 with about 5 ng per mg soluble protein. With regard to spleen concentrations, there emerged two high-expression (~5 µg/mg soluble protein), two low-expression, and two no-expression lines. Among the latter is line ML-310, with high boMx1 levels in the lungs and none in the spleen. Upon maximum stimulation by poly-I/C, fifth-passage embryonic fibroblasts from ML-549 and ML-555 mice expressed about 1.4 and 0.2 µg boMx1 per mg of soluble protein. Our stably transfected Vero cell line, which shows a high degree of VSV resistance (Baise et al. 2004), contains about 1 µg/mg, and the bovine BT and MDBK cell lines produce ~1 and ~0.6 µg/mg, respectively (Fig. 2D). For comparison we also conducted a large-scale screening of the boMx1 content of bovine spleens collected at a local

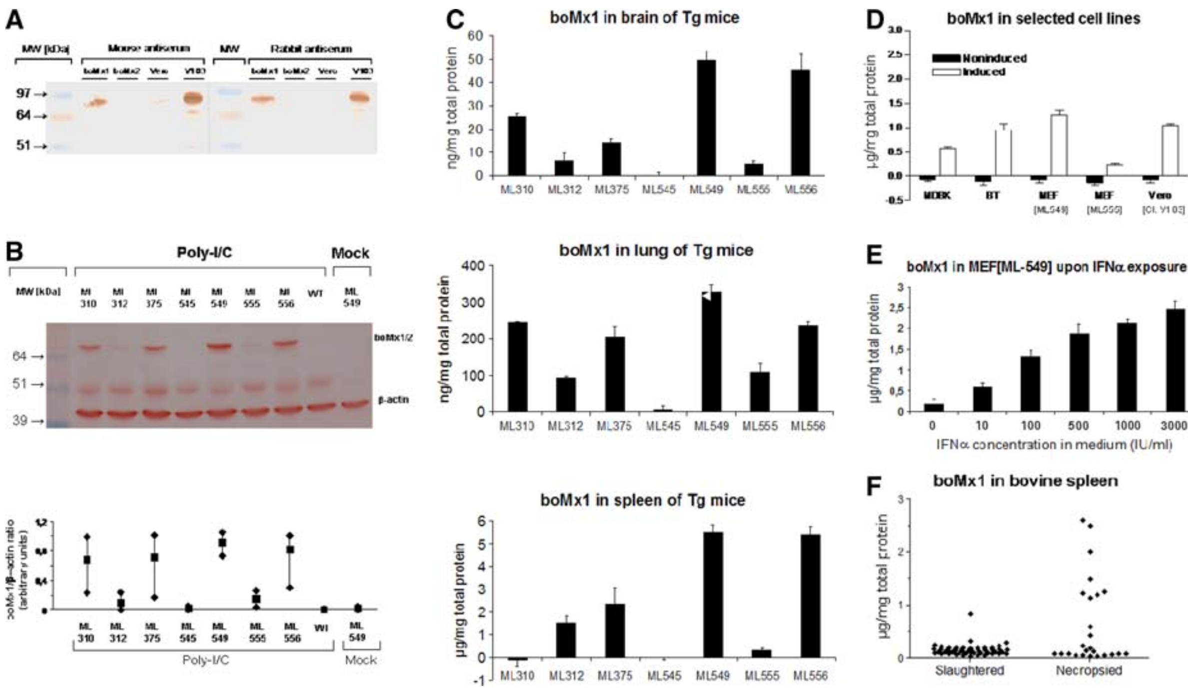
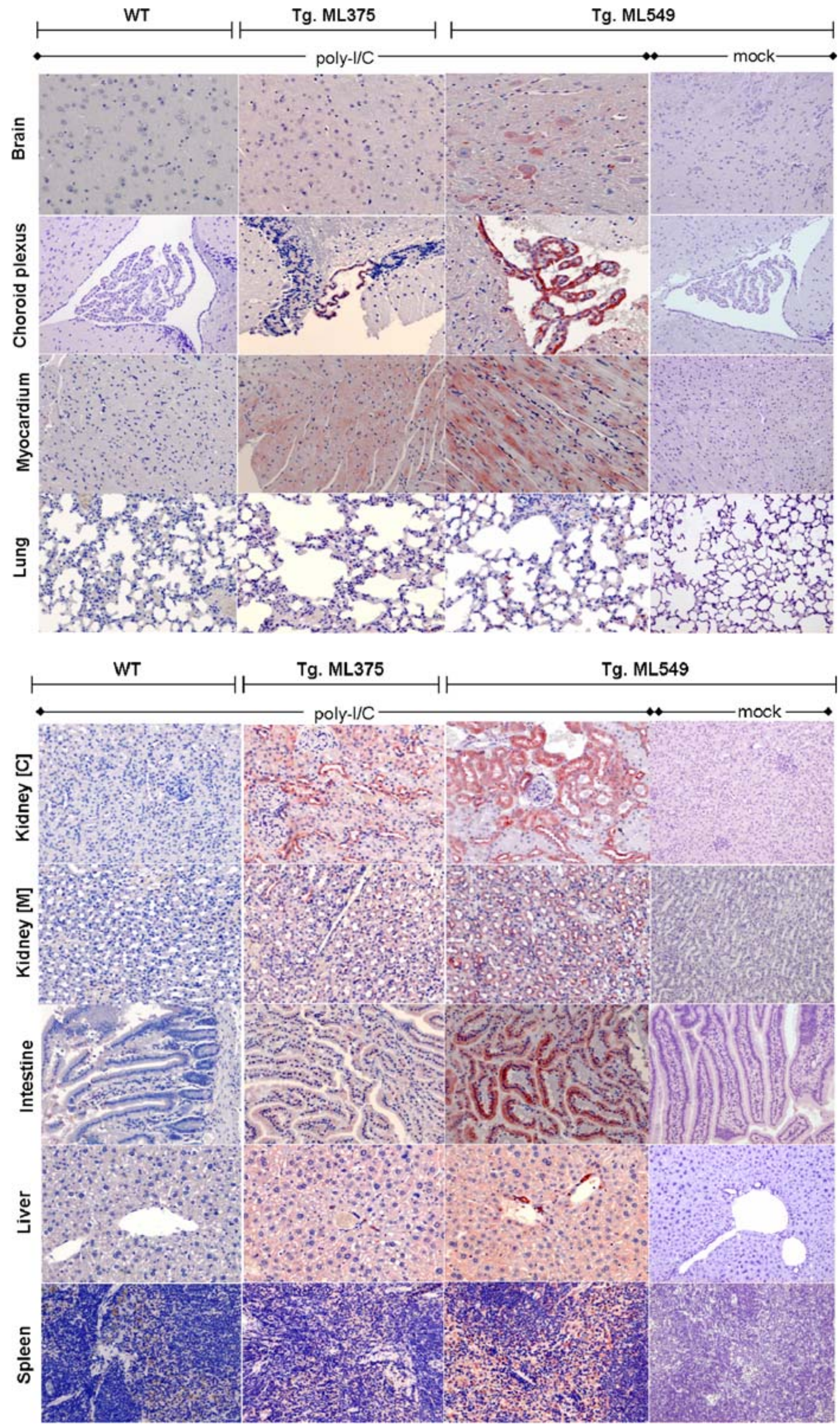


Fig. 2 Patterns of boMx1 expression among transgenic lines and comparison with the boMx1 content of bovine cells/tissues. **A** Mouse and rabbit antisera tag boMx1, but not boMx2. Vero cells were transiently transfected with cDNA constructs encoding boMx1, or boMx2. Vero cells and vero cells stably transfected with cDNA conditionally expressing the bovine *Mx1* gene are included (clone #V103, see Baise et al. 2004). About 24 h after transfection/stimulation, bovine Mx proteins were detected after 10% polyacrylamide SDS–PAGE of cell extracts and western blotting with rabbit (*left*) or mouse (*right*) antiserum. Transient expression of boMx2 had been duly detected by real time PCR. **B** Bovine Mx proteins are readily detectable in lung extracts from poly-I/C-pretreated transgenic mice, but not in extracts from either mock- or poly-I/C-pretreated ML-549 and wild-type mice, respectively. About 24 h after pretreatment, boMx1 protein and β-actin were detected in pooled lung protein extracts by 10% polyacrylamide SDS–PAGE and western blotting with mouse anti-boMx1 antiserum and an anti-actin mAb (*top*). Molecular weight markers are shown on the *left*. Densitometric semi-quantitation of the boMx1-associated bands was performed and the results reported are mean (squares) and extreme (lozenges) values retrieved from three independent immunoblots (*bottom*). **C** BoMx1 content determinations in brain, lung, and spleen from transgenic mice reveal line-specific expression patterns. BoMx1 values were determined in triplicate by ELISA on pooled ($n = 5$) protein extracts retrieved from organs taken from 8-week-old female mice that had been injected with poly-I/C 24 h earlier. See text for ELISA specifications. Values are means ± SD. **D** Spontaneous or forced boMx1 expression in bovine cells yields concentrations of the same order of magnitude as measured in transgenic mouse and Vero cells. Control and induced boMx1 expression in bovine cell lines (MDBK and BT), mouse embryonic fibroblasts (MEF) derived from transgenic lines ML-549 and ML-555, and in the Vero cell clone #V103. Values are means ± SD of triplicate ELISA determinations. Induction was done by adding poly-I/C (50 µg/ml) to the cell culture broth 24 h before harvest. **E** boMx1 expression is induced by exposure to IFNα and its level of expression is subordinated to IFNα concentration in the medium. Mouse embryonic fibroblasts from the ML-549 line were seeded in 24-well plates and IFNα was incorporated when the monolayers reached confluence and the reaction was stopped 24 h after. Values are means ± SD of duplicate ELISA determinations from protein extracts of each of three wells treated separately. **F** Spontaneous boMx1 expression in bovine spleens collected at a local slaughterhouse ($n = 80$) and at the Faculty necropsy clinic ($n = 25$). *Postmortem* spleen boMx1 concentrations amounted between 0.5 and 3 µg/mg among the ten virus-positive cases detected in the cohort of necropsied animals, thus being of the same order of magnitude as measured in transgenic mouse spleens

slaughterhouse and at the Faculty necropsy clinic. This revealed spontaneous (thus submaximal) concentrations of ~0.15 µg/mg soluble protein among slaughtered animals and among necropsied animals in which no viruses were detected, whereas boMx1 concentrations amounted between 0.5 and 3 µg/mg among the ten virus-positive cases. In summary, we have produced transgenic mice that lack endogenous antiviral Mx proteins (their genetic background is that of the FVB/J strain) but that conditionally express the bovine *Mx1* and *Mx2* genes, in various organs and under the control of their natural promoter,


WT
Tg. ML375
Tg. ML549
poly-I/C
mock
Brain
Choroid plexus
Myocardium
Lung
WT
Tg. ML375
Tg. ML549
poly-I/C
mock
Kidney [C]
Kidney [M]
Intestine
Liver
Spleen

◀ **Fig. 3** Immunohistochemical BoMx1 detection in tissues from transgenic mice reveals widespread expression along with line- and organ-specific patterns. Overall staining is always more intense in kidney and intestine than elsewhere. Intra-organ staining differences are also obvious. Staining is much more intense, for instance, in the choroid plexuses or in the Kupffer cells of the liver, and in the type II pneumocytes of the lungs than in the surrounding tissues. About 24 h before euthanasia/sampling, both wild-type and transgenic mice were induced by injection of either poly-I/C (15 μg/g body weight, ip) or saline

up to protein concentrations comparable to those measured in bovine cells and tissues.

Transgenic mice expressing the intact *Bos taurus* Mx genes are protected against lethal VSV infection

Using weight loss as a measure of morbidity, we then examined whether the bovine Mx genes could protect transgenic mice against lethal VSV infection. We observed that at $\sim 10^7$ $CCID_{50}$ of the virus, transgenic boMx$^{+/-}$ ML-549 mice did not experience any significant weight loss, whereas transgenic boMx$^{+/-}$ ML-555 and wild-type mice lost significant body weight after inoculation (Fig. 4A). Follow-up of survival among these mice revealed highly significant differences between ML-549 mice, on the one hand, and ML-555 and wild-type mice on the other (Kaplan-Meier analysis, $P < 0.01$); all ML-549 mice survived whereas the mortality rate was 100 and 70% among the two other lines, respectively (Fig. 4B). In summary, mice expressing both bovine Mx1 and Mx2 were protected against the high mortality and morbidity caused by the VSV virus. To test hypothesis that this boMx1/2-induced reduction in clinical severity was associated with repression of the virus itself, we quantified VSV viral loads by qPCR and conventional titration. The 4 days post-VSV infection, VSV genomic

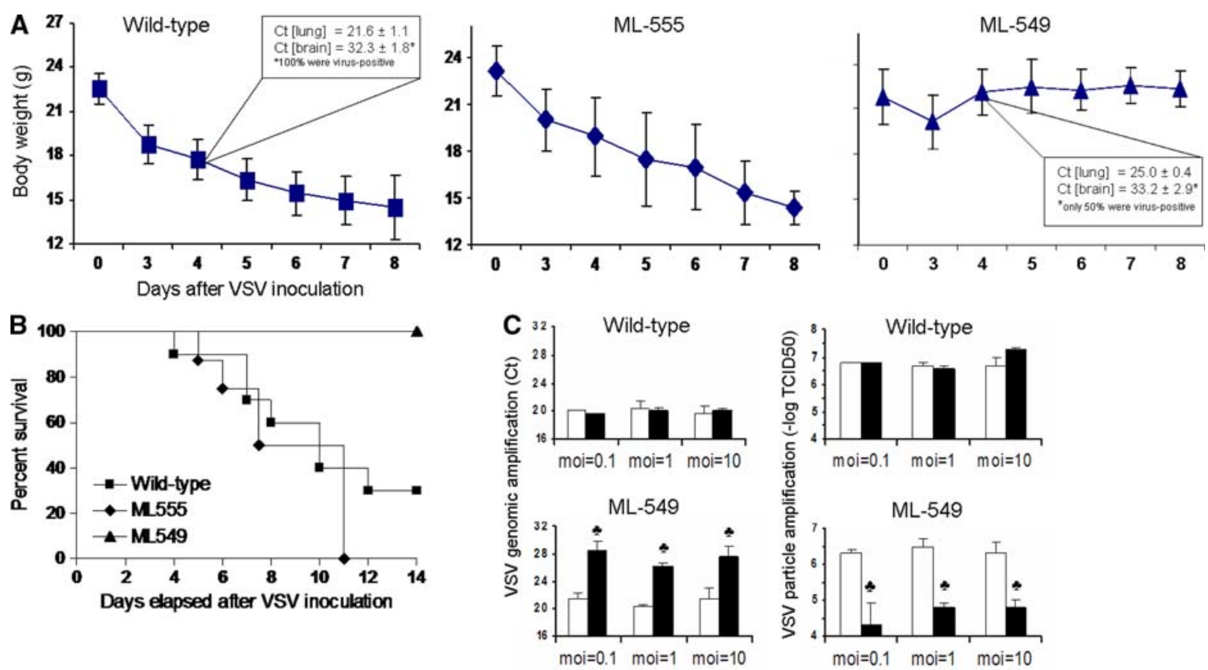


Fig. 4 Vesicular stomatitis virus (*VSV*, strain Indiana) is not lethal to high-expression boMx$^{+/-}$ (ML-549) mice at a dose lethal to both low-expression boMx$^{+/-}$ (ML-555) and zero-expression boMx$^{-/-}$ (wild-type) mice. **A** Body weight change (mean ± SD) and **B** survival of mice ($n = 10$ per group) inoculated with $\sim 10^7$ $CCID_{50}$. Inset: cycle thresholds (means ± SD) recorded for VSV genomic amplification in lung and brain homogenates from boMx$^{-/-}$ (wild-type) and boMx$^{+/-}$ (ML-549) mice 4 days after inoculation. **C** Effect of transgenic expression of the *Bos Taurus* Mx system on VSV genomic (*left*) and particle (*right*) amplification in embryonic fibroblasts derived from wild-type and transgenic (ML-549) mice, as measured on supernatants collected 24 h after virus incorporation. Values are means ± SD of two independent experiments. *White* and *black boxes* refer, respectively to mock- and IFNα-exposed cells. *Black clovers* point to results significantly different from those obtained with the corresponding mock-exposed cells ($P < 0.01$). m.o.i., multiplicity of infection

loads were significantly higher in the lungs of wild-type mice than in ML-549 mice, as judged from the increased cycle threshold in samples from the latter (Fig. 4A, inset). The brain is another target organ for VSV following intranasal infection (Chesler and Reiss 2002). On day 4 post infection, the VSV genome was retrieved from 100% of the wild-type but only 50% of the ML-549 mice, and a comparison of the qPCR-positive samples revealed a lower level in ML-549 mice. We also measured replication of VSV in the lung and brain tissues of the mice 4 days after inoculation. The virus titers found in the brains of all mice were below the limit of detection of our assay at this time point. The virus was detected in the lungs of all wild-type mice (4.17 ± 0.65 [-1]log$TCID_{50}$), but never in the lungs of the transgenic mice. Overall, expression of the bovine *Mx* genes in mice was thus crucial to reducing the viral load in lungs after VSV infection. About 24 h after infection of embryonic fibroblast monolayers with VSV, both the genome copy number and the infectious particle load were again dramatically lower in ML-549 than in wild-type-derived cells (Fig. 4C).

Discussion

We have created transgenic mouse lines in which the genomic fragment encoding the bovine antiviral Mx system is inserted and is transmitted from one generation to the next. As the genetic background of these lines is that of strain FVB/J, carrying defective Mx genes that no longer code for a functional endogeneous Mx protein (Fig. 1A, B), these transgenic lines constitute an ideal experimental model for testing the antiviral activities of bovine Mx proteins in vivo. The FAM3B gene included in the BAC encodes a 235-amino-acid cytokine with a secretion signal peptide. As this cytokine is expressed specifically in the islets of Langerhans of the pancreas and the testes (Zhu et al. 2002), its presence, if any, is unlikely to generate any artefact when the antiviral function of the bovine Mx proteins is examined.

Several attempts of this type have been made in the past to examine the antiviral function of the mouse Mx1 protein (Arnheiter et al. 1990; Kolb et al. 1992; Müller et al. 1992) or the human MxA protein (Pavlovic et al. 1995) in vivo. These previous attempts have always consisted in inserting a small artificial construct bearing a mouse promoter (that of Mx1, 3-hydroxy-3-methylglutaryl coenzyme A reductase, or albumin), a human promoter (that of metallothioneine IIA), or a viral promoter (the SV40 early enhancer/promoter) upstream from the *moMx1* or *huMxA* cDNA. Many expression problems appeared: absence of transcription (Pavlovic et al. 1995), absence of mRNA translation (Müller et al. 1992), gradual extinction (Kolb et al. 1992), mosaicism (Arnheiter et al. 1990; Kolb et al. 1992), or loss of inducibility of expression (Arnheiter et al. 1990). To avoid these problems we opted for a large transgene ($\sim$210 kb) because we could legitimately expect it to contain both genes in their entirety, including the short-range (UTRs, introns) and long-range *cis*-regulatory elements required for correct spatio-temporal expression. The second advantage of the use of a BAC is that BACs are expected to be more resistant to position effects than smaller transgenes (Giraldo and Montoliu 2001; Gong et al. 2003). Overall, we have confirmed the advantages linked to a larger transgene size, since we have detected inducible production of the boMx proteins in all organs of $\sim$90% (8/9) of the lines showing germinal insertion, whereas expression of the huMxA protein was previously demonstrated in only $\sim$13% (2/15) of the transgenic lines obtained (Pavlovic et al. 1995). The transgenic lines produced here differ slightly between each other as regards the regulation of both transcription and translation. Firstly, the levels of mRNA measured after standardized induction and the ratio of *boMx1* to *boMx2* transcripts were found to vary from line to line. For example, the *boMx2* transcript appears to be totally absent from ML-555 (Fig. 1B). On the other hand, a look at transcript levels in relation to the corresponding protein concentrations suggests that the rate of conversion from transcripts to protein also varies from line to line. Furthermore, we have found expression to vary considerably from organ to organ (spleen > lung > brain), with a clear variation of the spatial expression pattern according to the line. Expression differences between lines thus appear to be slighter in the lungs and greater in the spleen, as illustrated by the spectacular case of line ML-310, which shows high expression in the lungs and zero expression in the spleen. These differences probably reflect different numbers of insertions and/or random excision of certain BAC fragments in the course of purification and pronuclear microinjection. The absence of *boMx2* transcripts in line ML-555 suggests insertion of a single, truncated BAC. It is likewise plausible that the other differences stem from excision of certain remote *cis*-regulatory elements. Similarly, the fact that the two high-expression lines share a common spatial expression pattern may reflect insertion of several BACs, complete or truncated, ensuring the presence of at least one copy of all the regulatory elements present in nature.

As the objective was to validate the use of these transgenic mice in characterizing the functions of the two bovine Mx proteins in vivo, it was important to examine whether the protein concentrations obtained are biologically pertinent or not. Our comparison of maximum expression levels measured after poly-I/C induction in bovine and transgenic-mouse-derived cell lines shows that the amount of boMx1 accumulated over a 24 h period in response to poly-I/C was practically the same in bovine MDBK and BT cells, in embryonic fibroblasts from the high-expression line ML-549, and in cells of clone Vero #V103, where the antiviral functions of the boMx1 protein were previously studied (Baise et al. 2004; Leroy et al. 2005, 2006): roughly between 0.6 and 1.4 μg/mg soluble protein (Fig. 2D). The level recorded in embryonic fibroblasts of the low-expression line ML-555 reached about a third/quarter of the level recorded in MDBK and BT bovine cells, respectively. These results indicate that insertion of the chosen BAC transgene has enabled us to generate two high-expression lines that both faithfully reproduce the conditional expression pattern typical of the species from which the transgene derived and produce similar quantities of boMx1. We have additionally compared the levels of boMx1 reached in the spleens of a cohort of bovines in response to a viral disease with the concentrations generated artificially in response to poly-I/C in the spleens of transgenic mice (Fig. 2D). Again the concentrations arising spontaneously in the spleens of infected bovines was of the same order of magnitude as the maximal concentrations measured in the spleens of transgenic mice. This confirms that the Mx system introduced into the transgenic mice functionally mimics the Mx system present in the species of origin, *Bos taurus*.

We further show that a high degree of resistance to VSV can be conferred to mice by germinal transformation. This resistance should reflect both the effective antiviral action of the protein(s) concerned and its/their inducibility by the virus itself. We conclude that the boMx1 protein, the boMx2 protein, or both and their regulatory machinery constitute a genetic substrate capable of transforming an inexorably fatal infection into a benign and transient one (Fig. 4A, B). This confirms in vivo the anti-VSV activities detected in vitro, here in embryonic fibroblasts from line ML-549 (Fig. 4C) and previously in Vero cells for boMx1 (Baise et al. 2004) and in NIH-3T3 cells for boMx2 (Babiker et al. 2007). As the viral disease is not attenuated at all in line ML-555 despite production of boMx1, one might hypothesize that the protection conferred in vivo requires the simultaneous presence of both boMx1 and boMx2. Alternatively, the protection conferred in vivo might be dose-dependent, since an ~3-fold-lower level of boMx1 is observed in the lungs and an ~10-fold-lower level in the brain in line ML-555 (Fig. 3C). A third possibility is that the time required for the antiviral protein to accumulate to an effective level is crucial. This timing is likely to depend on the presence or absence of as yet unidentified regulatory elements that line ML-555 might lack.

In conclusion, we have generated transgenic mice expressing the complete bovine antiviral Mx system under the control of its close and remote natural regulatory elements. The various transgenic lines produced differ as regards their spatio-temporal *Mx* expression patterns, which suggests that predictable breaks in the BAC used for pronuclear injection may have caused amputation of certain remote elements in certain lines. We have further shown that two of these lines display boMx1 protein levels quite similar to those observed in bovine tissues, and that transgenesis has conferred to these lines a high degree of resistance to VSV amplification and VSV-associated disease. These transgenic lines constitute good models for identifying new regulatory elements and for studying in vivo the various known and as yet undiscovered antiviral activities of the bovine Mx proteins, with priority emphasis on the exceptional antirabic function of boMx1 recently detected in vitro.

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